6.3: Famous Sums of Continuous Random Variables

When finding the density of common sums, we ultimately stumble across two very interesting examples. The first of these is the sum of two *independent* normal random variables.

Let X and Y be independent normal random variables with means μ_X, μ_Y and variances σ_X^2, σ_Y^2 respectively. Prove that $Z = X + Y$ is normal, and find its mean and variance

ASSUME $X = N(0, a)$ $Y = N(0, b)$

$$f_Z(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t-x)\,dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi a}} e^{-\frac{x^2}{2a}} \cdot \frac{1}{\sqrt{2\pi b}} e^{-\frac{(t-x)^2}{2b}}\,dx = \frac{1}{2\pi\sqrt{ab}} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2a} - \frac{(t-x)^2}{2b}}\,dx$$

EXPONENT:

$$\frac{-x^2}{2a} - \frac{(t-x)^2}{2b} = \frac{-bx^2 - a(t^2 - 2tx + x^2)}{2ab} = \frac{\overbrace{-(a+b)x^2}^{-w^2} + 2atx - at^2}{2ab}$$

$w = x \cdot \sqrt{a+b}$

$dw = \sqrt{a+b} \cdot dx$

$$= \frac{-w^2 + 2at \cdot \frac{w}{\sqrt{a+b}} - at^2}{2ab} = \frac{-\left(w^2 - \frac{2at}{\sqrt{a+b}} \cdot w + \left(\frac{at}{\sqrt{a+b}}\right)^2\right) - at^2 + \left(\frac{at}{\sqrt{a+b}}\right)^2}{2ab}$$

$$= \frac{-\left(w - \frac{at}{\sqrt{a+b}}\right)^2}{2ab} + \frac{-at^2 + \frac{a^2t^2}{a+b}}{2ab} \Rightarrow \frac{-a^2t^2 - abt^2 + a^2t^2}{2ab(a+b)} = \frac{-t^2}{2(a+b)}$$

$N\left(\frac{at}{\sqrt{a+b}}, ab\right)$

$$= \frac{1}{2\pi\sqrt{ab}} \int_{-\infty}^{\infty} e^{\frac{-\left(w - \frac{at}{\sqrt{a+b}}\right)^2}{2ab}} \cdot e^{\frac{-t^2}{2(a+b)}} \cdot \frac{dw}{\sqrt{a+b}} = \frac{1}{\sqrt{2\pi(a+b)}} e^{-\frac{t^2}{2(a+b)}} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi ab}} e^{\frac{-\left(w - \frac{at}{\sqrt{a+b}}\right)^2}{2ab}}\,dw}_{1}$$

$$= \frac{1}{\sqrt{2\pi(a+b)}} e^{-\frac{t^2}{2(a+b)}} \longleftarrow \text{DENSITY OF } N(0, a+b)$$

$N(0,a) + N(0,b) = N(0, a+b)$

$\tilde{X} = \mu_X + X$

$\tilde{Y} = \mu_Y + Y$

$\tilde{Z} = \tilde{X} + \tilde{Y} = \mu_X + \mu_Y + X + Y = \mu_X + \mu_Y + Z = N(\mu_X + \mu_Y, a+b)$

This means that for any independent normal random variables $X_k = \mathcal{N}(\mu_k, \sigma_k^2)$ we know that

$$\sum_{k=1}^{n} X_k = \mathcal{N}\left(\sum_{k=1}^{n} \mu_k, \sum_{k=1}^{n} \sigma_k^2\right)$$

Example: Every month Rich gets his allowance deposited to his bank account. The amount is a normally distributed random variable with mean \$1000 and variance \$300. If he doesn't touch it for a year, what is the probability that he has more than \$12300 after one year?

MONTH k DEPOSIT $= X_k = N(1000, 300)$

YEAR TOTAL $= Y = X_1 + \cdots + X_{12} = \sum_{k=1}^{12} X_k = N\left(\sum_{k=1}^{12} 1000, \sum_{k=1}^{12} 300\right) = N(12000, 3600)$

$P(Y > 12300) = P(12000 + \sqrt{3600} \cdot Z > 12300)$

$= P(60 \cdot Z > 300)$

$= P(Z > 5) = 1 - P(Z \leq 5)$

$\boxed{= 1 - \Phi(5)} = 0.00000029$

Example: The height of the average American man is 69 inches with a variance of 9. The height of the average American woman is 65 inches with a variance of 7. What is the probability that a randomly selected American woman is taller than a randomly selected American man?

$M = N(69, 9)$

$W = N(65, 7)$

$M - W \neq N(69 - 65, 9 - 7)$

$\neq N(4, 2)$

$M - W = N(69 - 65, 9 + 7) = N(4, 16)$

(VARIANCES ALWAYS +)

$P(W > M)$

$= P(W - M > 0) = P(M - W < 0)$

$= P(4 + \sqrt{16} \cdot Z < 0)$

$= P(4Z < -4)$

$= P(Z < -1)$

$\boxed{= \Phi(-1)}$

$= 0.15865525$

Another common sum occurs when adding copies of exponential random variables.

Let $X_1, X_2, \ldots$ be independent copies of the exponential random variable $Exp(\lambda)$

Define $Y_n := \sum_{k=1}^{n} X_k$, what is the density of Y_n?

$$Y_1 = X_1 \longrightarrow f_{Y_1}(t) = f_{X_1}(t) = \lambda e^{-\lambda t} \quad (t \geq 0)$$

$$Y_2 = X_1 + X_2 \longrightarrow f_{Y_2}(t) = f_{X_1} * f_{X_2} = \int_{-\infty}^{\infty} f_{X_1}(x) f_{X_2}(t-x)\,dx = \int_0^t \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(t-x)}\,dx$$

$$= \lambda^2 \int_0^t e^{-\lambda t}\,dx = \lambda^2 e^{-\lambda t} \int_0^t dx = \lambda^2 e^{-\lambda t} \cdot x \Big|_0^t = \lambda^2 t e^{-\lambda t} \quad (t \geq 0)$$

$$Y_3 = X_1 + X_2 + X_3 = Y_2 + X_3 \longrightarrow f_{Y_3}(t) = f_{Y_2} * f_{X_3} = \int_{-\infty}^{\infty} f_{Y_2}(x) f_{X_3}(t-x)\,dx = \int_0^t \lambda^2 x e^{-\lambda x} \cdot \lambda e^{-\lambda(t-x)}\,dx$$

$$= \lambda^3 e^{-\lambda t} \int_0^t x\,dx = \lambda^3 e^{-\lambda t} \cdot \frac{x^2}{2}\Big|_0^t = \lambda^3 \frac{t^2}{2} e^{-\lambda t} \quad (t \geq 0)$$

$$f_{Y_n}(t) = \lambda^n \cdot \frac{t^{n-1}}{(n-1)!} e^{-\lambda t} \quad (t \geq 0)$$

This may look familiar if you have heard of the **gamma function**: $\Gamma(a) = \int_0^\infty t^{a-1} e^{-t}\,dt$

A quick proof by induction shows us $\Gamma(n) = (n-1)!$ for all $n = 1, 2, \ldots$

$n=1$? $\quad \Gamma(1) = \int_0^\infty t^{1-1} e^{-t}\,dt = \int_0^\infty e^{-t}\,dt = -e^{-t}\Big|_0^\infty = (0 - (-1)) = 1 = 0!$

$$\Gamma(n+1) = \int_0^\infty t^{(n+1)-1} e^{-t}\,dt = \int_0^\infty t^n e^{-t}\,dt \quad \begin{pmatrix} u = t^n & dv = e^{-t}dt \\ du = nt^{n-1}dt & v = -e^{-t} \end{pmatrix}$$

$$= \underbrace{t^n \cdot (-e^{-t})\Big|_0^\infty}_{0-0} - \int_0^\infty -e^{-t}(n t^{n-1}\,dt) = n \int_0^\infty t^{n-1} e^{-t}\,dt$$

$$\Gamma(n+1) = n \cdot \Gamma(n)$$

INDUCTIVE STEP: IF $\Gamma(n) = (n-1)!$

$$\Gamma(n+1) = n \cdot (n-1)! = n! = ((n+1)-1)!$$

Which leads us to the standard gamma distribution, a variable $X = Gam(\gamma)$ with $f_X(t) = \frac{t^{\gamma-1}e^{-t}}{\Gamma(\gamma)}$ on $[0, \infty)$

Find the mean and variance of $X = Gam(\gamma)$

$$E[X] = \int_{-\infty}^{\infty} t f_X(t)\,dt = \int_0^{\infty} t \cdot \frac{t^{\gamma-1}e^{-t}}{\Gamma(\gamma)}\,dt = \frac{1}{\Gamma(\gamma)} \int_0^{\infty} t^{(\gamma+1)-1} e^{-t}\,dt = \frac{\Gamma(\gamma+1)}{\Gamma(\gamma)} = \frac{\gamma\Gamma(\gamma)}{\Gamma(\gamma)} = \boxed{\gamma}$$

$$E[X^2] = \int_{-\infty}^{\infty} t^2 f_X(t)\,dt = \int_0^{\infty} t^2 \frac{t^{\gamma-1}e^{-t}}{\Gamma(\gamma)}\,dt = \frac{1}{\Gamma(\gamma)} \int_0^{\infty} t^{(\gamma+2)-1} e^{-t}\,dt = \frac{\Gamma(\gamma+2)}{\Gamma(\gamma)} = \frac{(\gamma+1)\cdot\gamma\Gamma(\gamma)}{\Gamma(\gamma)} = \gamma^2+\gamma$$

$$Var(X) = E[X^2] - (E[X])^2 = \gamma^2+\gamma - (\gamma)^2 = \boxed{\gamma}$$

The gamma function shows up in many integrals, but one particularly interesting one is:

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

This leads us to another famous distribution called the **Beta** distribution (with parameters a,b).

The random variable X with density $f_X(t) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} t^{a-1}(1-t)^{b-1}$ on $[0, 1]$ is $X = Beta(a, b)$

Find the mean and variance of $X = Beta(a, b)$

$$E[X] = \int_{-\infty}^{\infty} t \cdot f_X(t)\,dt = \int_0^1 t \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} t^{a-1}(1-t)^{b-1}\,dt = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 t^{(a+1)-1}(1-t)^{b-1}\,dt$$

$$= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \cdot \frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+1+b)} = \frac{\Gamma(a+1)}{\Gamma(a)} \cdot \frac{\Gamma(a+b)}{\Gamma(a+b+1)} = a \cdot \frac{1}{a+b} \boxed{= \frac{a}{a+b}}$$

$$E[X^2] = \int_0^1 t^2 \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} t^{a-1}(1-t)^{b-1}\,dt = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 t^{(a+2)-1}(1-t)^{b-1}\,dt$$

$$= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \cdot \frac{\Gamma(a+2)\Gamma(b)}{\Gamma(a+2+b)} = \frac{\Gamma(a+2)}{\Gamma(a)} \cdot \frac{\Gamma(a+b)}{\Gamma(a+b+2)} = (a+1)\cdot a \cdot \frac{1}{(a+b+1)(a+b)}$$

$$\boxed{Var(X) = \frac{(a+1)a}{(a+b+1)(a+b)} - \left(\frac{a}{a+b}\right)^2}$$

If $X = Gam(\gamma)$ and $Y = Gam(\delta)$ are independent, find the density of $X + Y$

$$f_{X+Y}(t) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(t-x)\, dx = \int_0^t \frac{1}{\Gamma(\gamma)} x^{\gamma-1} e^{-x} \cdot \frac{1}{\Gamma(\delta)} (t-x)^{\delta-1} e^{-(t-x)}\, dx$$

$$= \frac{1}{\Gamma(\gamma)\Gamma(\delta)} e^{-t} \int_0^t x^{\gamma-1} \cdot (t-x)^{\delta-1}\, dx$$

$$\left(\begin{array}{l} tw = x \\ t \cdot dw = dx \\ x = 0 \text{ --- } w = 0 \\ x = t \text{ --- } w = 1 \end{array} \right)$$

$$= \frac{1}{\Gamma(\gamma)\Gamma(\delta)} e^{-t} \cdot \int_0^1 (tw)^{\gamma-1} \cdot (t - tw)^{\delta-1} \cdot t\, dw$$

$$= \frac{1}{\Gamma(\gamma)\Gamma(\delta)} e^{-t} \cdot t^{\gamma-1} \cdot t^{\delta-1} \cdot t \underbrace{\int_0^1 w^{\gamma-1} (1-w)^{\delta-1} \cdot dw}_{\text{BETA INTEGRAL} \; = \; \frac{\Gamma(\gamma)\Gamma(\delta)}{\Gamma(\gamma+\delta)}}$$

$$= \frac{1}{\Gamma(\gamma)\Gamma(\delta)} e^{-t} \cdot t^{\gamma+\delta-1} \cdot \frac{\Gamma(\gamma)\Gamma(\delta)}{\Gamma(\gamma+\delta)}$$

$$= \frac{1}{\Gamma(\gamma+\delta)} t^{\gamma+\delta-1} e^{-t}$$

← DENSITY OF $\Gamma(\gamma+\delta)$